

Aarav Purohit

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EDUCATION

University of Southern California (USC), Los Angeles, CA

US Citizen

Data Science and AI undergraduate with hands-on experience in machine learning, cloud-based data pipelines, and client-facing analytics solutions.

Double-Major: Data Science; Cognitive Science (Computer Science focus)

Minor: Artificial Intelligence (AI)

Expected Graduation: May 2028 | GPA: 3.7

Awards: IB Scholar with Honors, AP Scholar with Distinction, 2nd place in UCLA Rover Engineering Design

Relevant Coursework: Applied Python, Data Structures and Algorithms, AI Principles and Foundations, Stats and Probability, Discrete Math in Computer Science, Embedded Systems, Statistics, Foundations of Data Management, Intro to Data Science, Front End Programming, Engineering Design

TECHNICAL SKILLS

Programming Languages: Python, SQL, C++, C, Java, JavaScript, HTML, CSS, MQL, Cypher

Libraries: Pandas, Pyspark, Scikit-Learn, PyTorch, Matplotlib, Seaborn, NumPy

Frameworks & Tools: AWS(Lambda, S3, Glue, Athena), Microsoft Power BI, SpringBoot, Maven, GitHub, SAS Viya, Microelectronics, Circuitry, IntelliJ, Oracle, Maven, VS Code, PyCharm, Node.js, Kaggle, Tableau, Mongo DB, NEO 4js.

EXPERIENCE

The Zenith Insurance Company

Woodland Hills, CA

AI and Business Technology Intern

(Jun 2025–Aug 2025, Dec 2025–Jan 2025)

- Designed a **Retrieval-Augmented Generation (RAG) chatbot** that processed company documents and enabled employees to query information **instantly**, severely **reducing** the need for manual search by employees
- Integrated secure login authentication and implemented prompt guardrails within the RAG chatbot, **ensuring 100% compliance** with company security and policy standards
- Collaborated with **10+ VPs** to source company documents, ensuring chatbot accuracy & coverage on all company functions
- Designed automated data ingestion and transformation workflows** to feed structured company data into LLM-based validation pipelines, improving document data accuracy by 30%, and identifying the most cost-effective solution.
- Worked alongside senior software engineers** to develop a company tool that identifies fraudulent claims and expedites claim processing

Java Developer Intern (Jun 2024–Sep 2024)

- Developed and implemented a Java-based solution to **standardize 10+ invoice formats**, as well as **cutting processing time by 20%**
- Designed a user authentication system in a Java Spring Boot application, **securing access for 100+ users** with a database
- Worked with senior developers to refine database querying techniques, **cutting authentication time** and improving system efficiency

Circuit Crafters' Club

Granada Hills Charter High School, CA

Founder & President

(Oct 2022 –May 2024)

- Founded and led a school-sponsored Microelectronics Club, **mentoring 30+ students** in circuit design and C++ programming, fostering hands-on learning and innovation **sparking student interest in Physics & Engineering**
- Designed and implemented **15+ advanced sensor-based projects**, such as harmful gas detectors, LCD displays, heartbeat monitors, and timer-based circuits, enhancing real-world application skills

eContek

Chatsworth, CA

IT Intern and Research Assistant

(Oct 2022–Dec 2023)

- Conducted research on **Artificial Intelligence (AI)** and **Machine Learning (ML)**, exploring advanced concepts such as Quantum AI and Machine Learning for 2-3 weeks at a time to stay ahead in emerging technologies
- Developed and implemented **Java** and **Python** programs to gain hands-on experience with **industry-standard libraries** and coding techniques, strengthening proficiency in real-world software development

PROJECTS

Climbing Performance Data Pipeline

Nov 2025-Present

- Designing a serverless **AWS data pipeline** to automate ingestion and transformation of climbing performance data using **Lambda, S3, and AWS Glue**, enabling scalable ETL automation and analytics-ready data storage.
- Implementing early stages of **ETL automation** for raw climb log processing and schema design for Athena queries
- Applying **data engineering principles** to build scalable, cloud-based analytics workflows, with **end-to-end cloud-based data**

NBA Game Winner Predictor

Mar 2025-Apr 2025

- Built a **Random Forest model** trained on NBA game results, player stats, and advanced team metrics (offensive/defensive ratings, win percentage, home-court advantage)
- Developed a **Flask-based web app** with an interactive UI displaying team colors, starting lineups, and stats alongside predictions.
- Gained experience in Machine Learning, data processing (**Pandas, Scikit-Learn**), and full-stack development (**Flask, HTML/CSS**).

LEADERSHIP AND INVOLVEMENT

Cyber Patriots

Granada Hills, CA

Officer-Social Media Outreach

Sep 2022–May 2024

- Competed in the National Youth Cyber Education Program, created by the Air Force Association, engaging in year-long cybersecurity challenges to **resolve simulated cyberattacks on Windows systems**
- Top 50** out of **3,000+ high schools statewide**, demonstrating exceptional teamwork and technical skills in securing operating systems